A statisticians meandering in climate science

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UNC Chapel Hill Nov ’26

Who Am I?

- 2007 UG: Mathematics with statistics
- 2008 MSc: Medical Statistics
- 2012 PhD with Shell
- 2011 Post-doc with Shell + Garrad Hassan
- 2013 Assistant Professor
- 2018 Associate Professor
- 2022 Professor
- 2026 Moved to the US

Maternity leave in 2013 and 2015. Long-term sick leave in 2022.

Interesting things?

- Developing methods for analyzing data whose statistical properties change over time.

Interesting things?

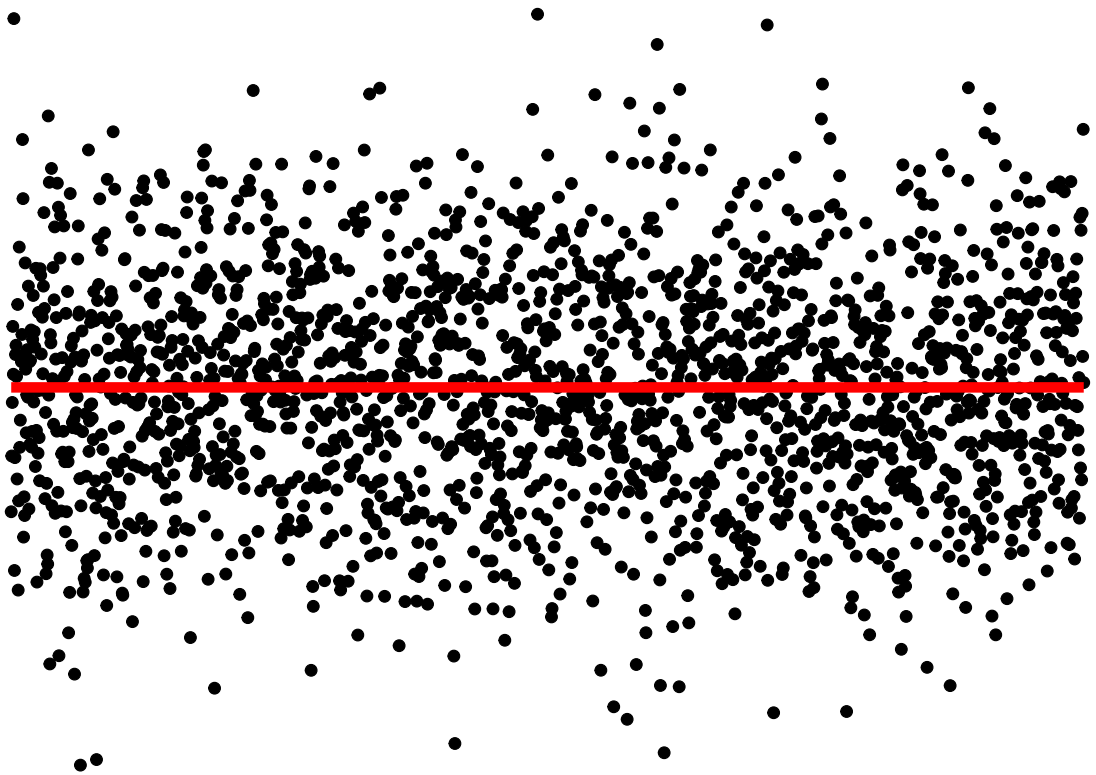
- Developing methods for analyzing data whose statistical properties change over time.

But what does that mean?

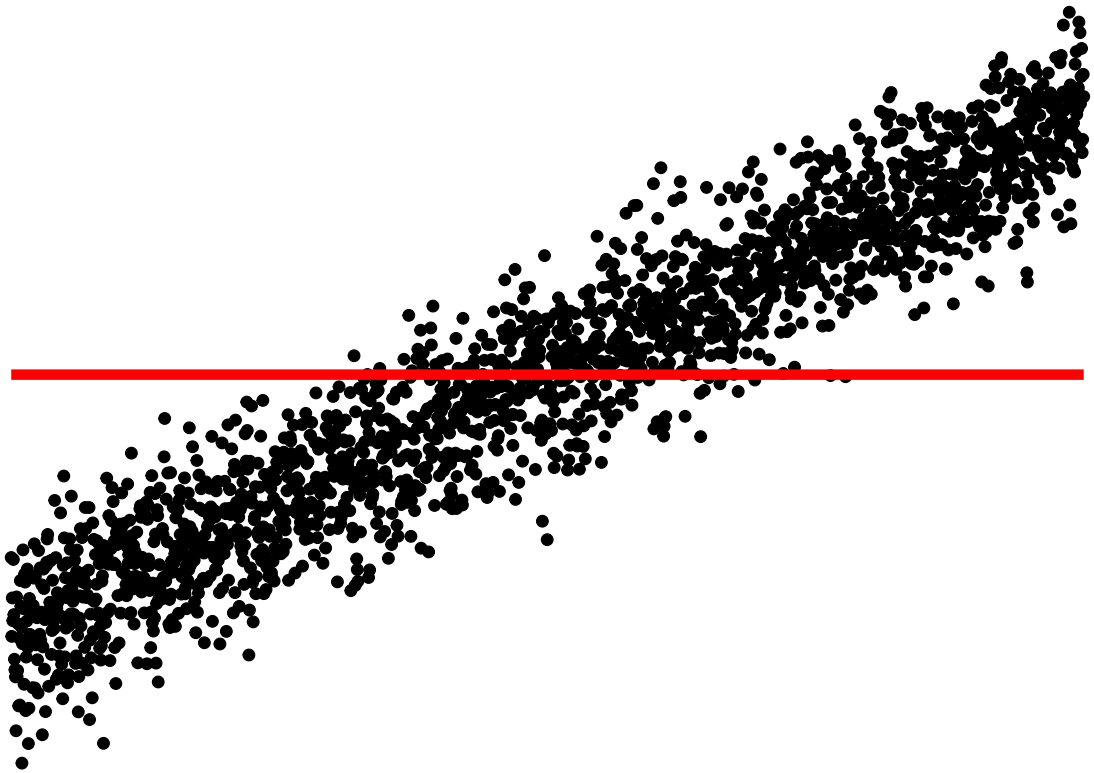
Interesting things?



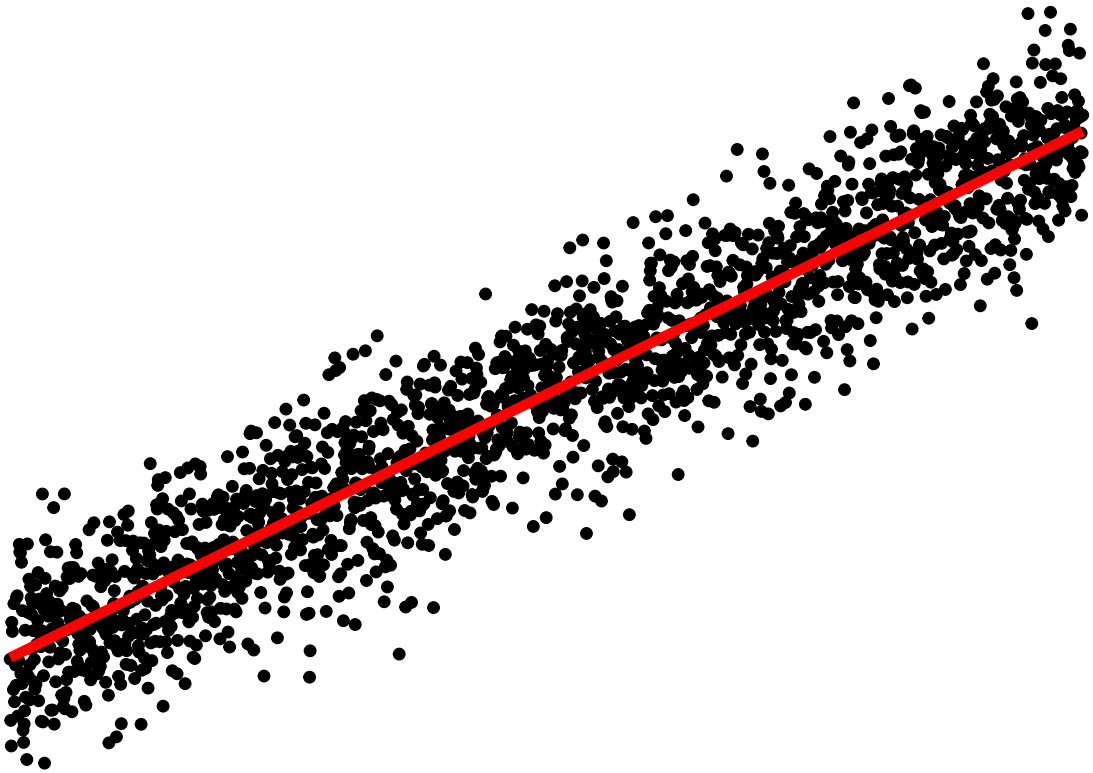
Model development



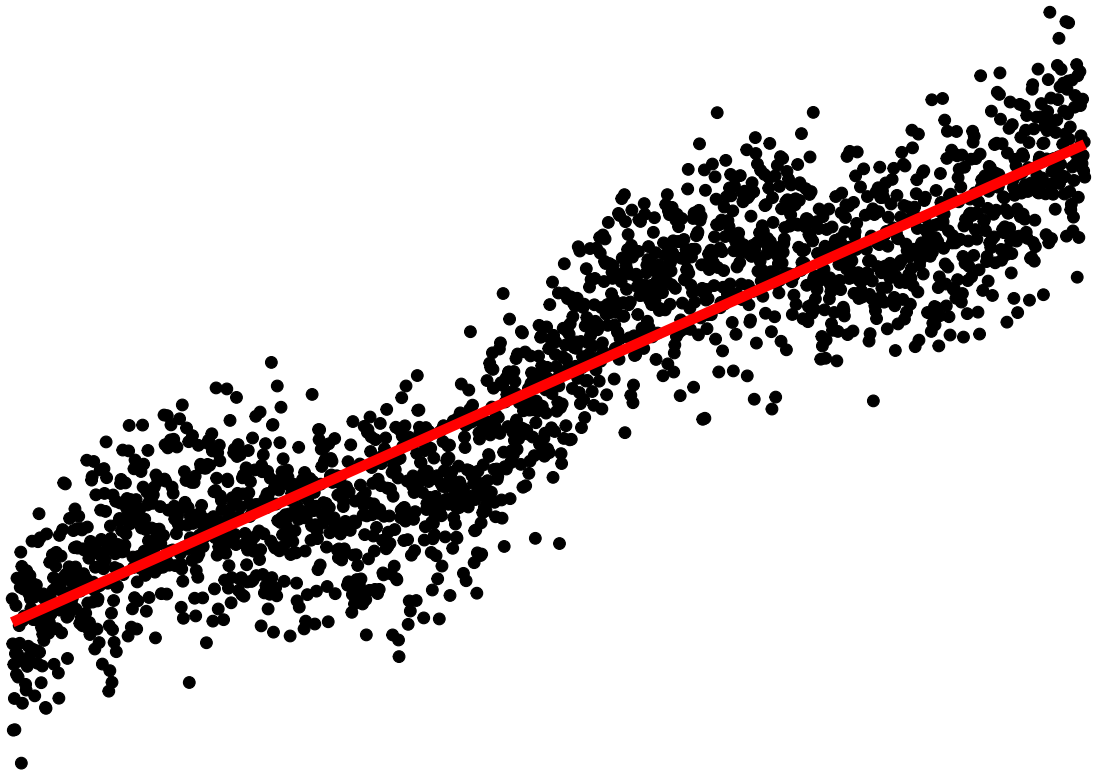
Model development



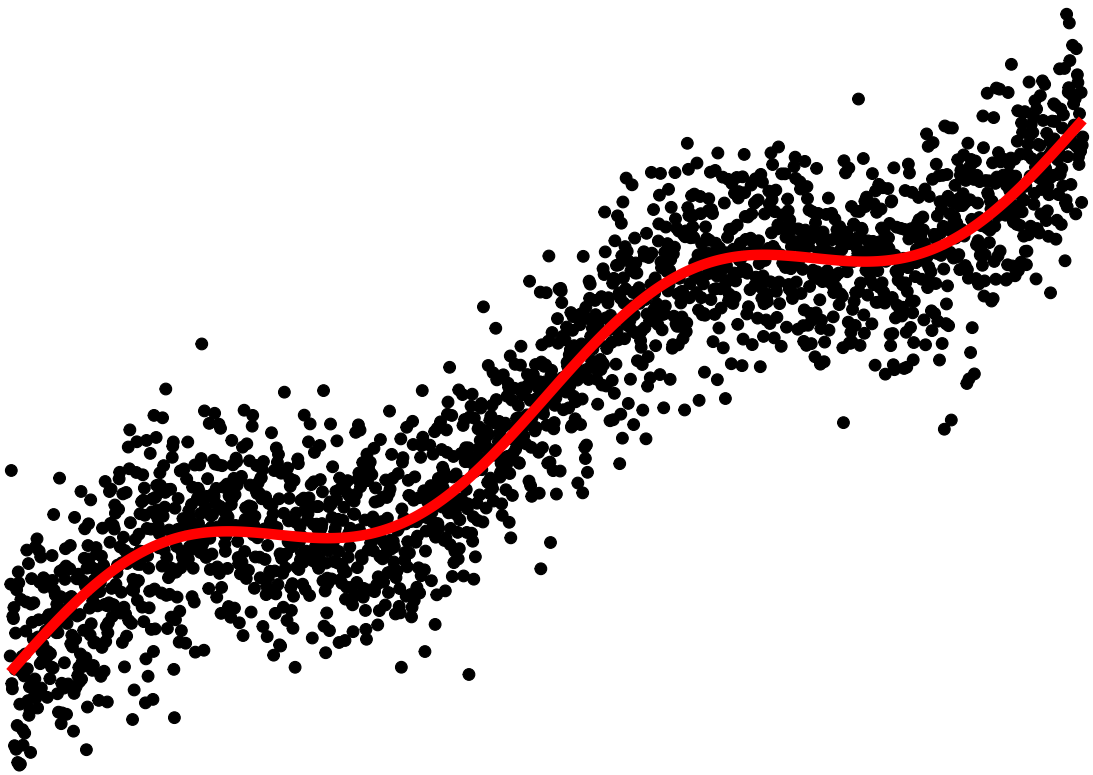
Model development



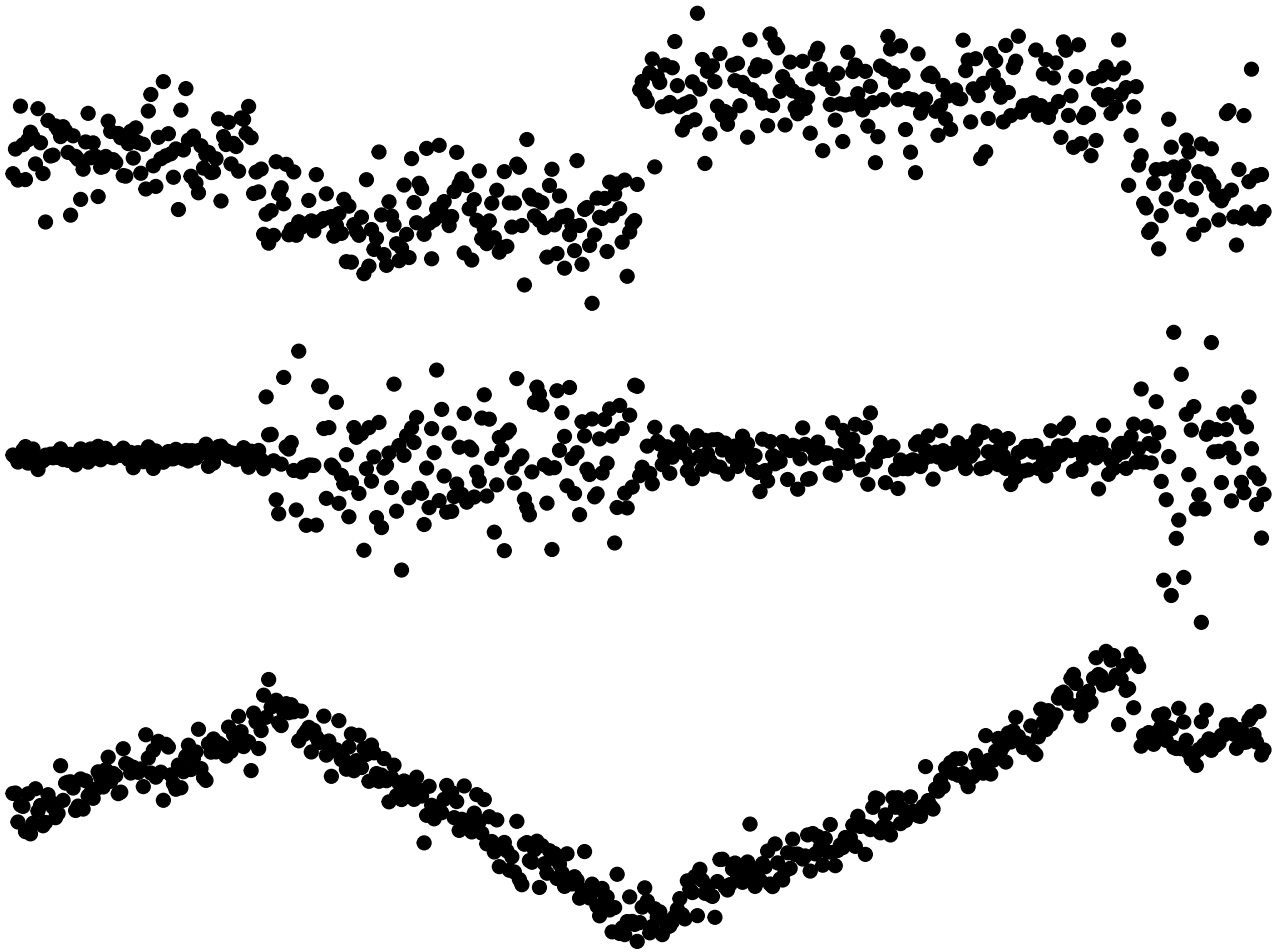
Model development



Model development



Model development



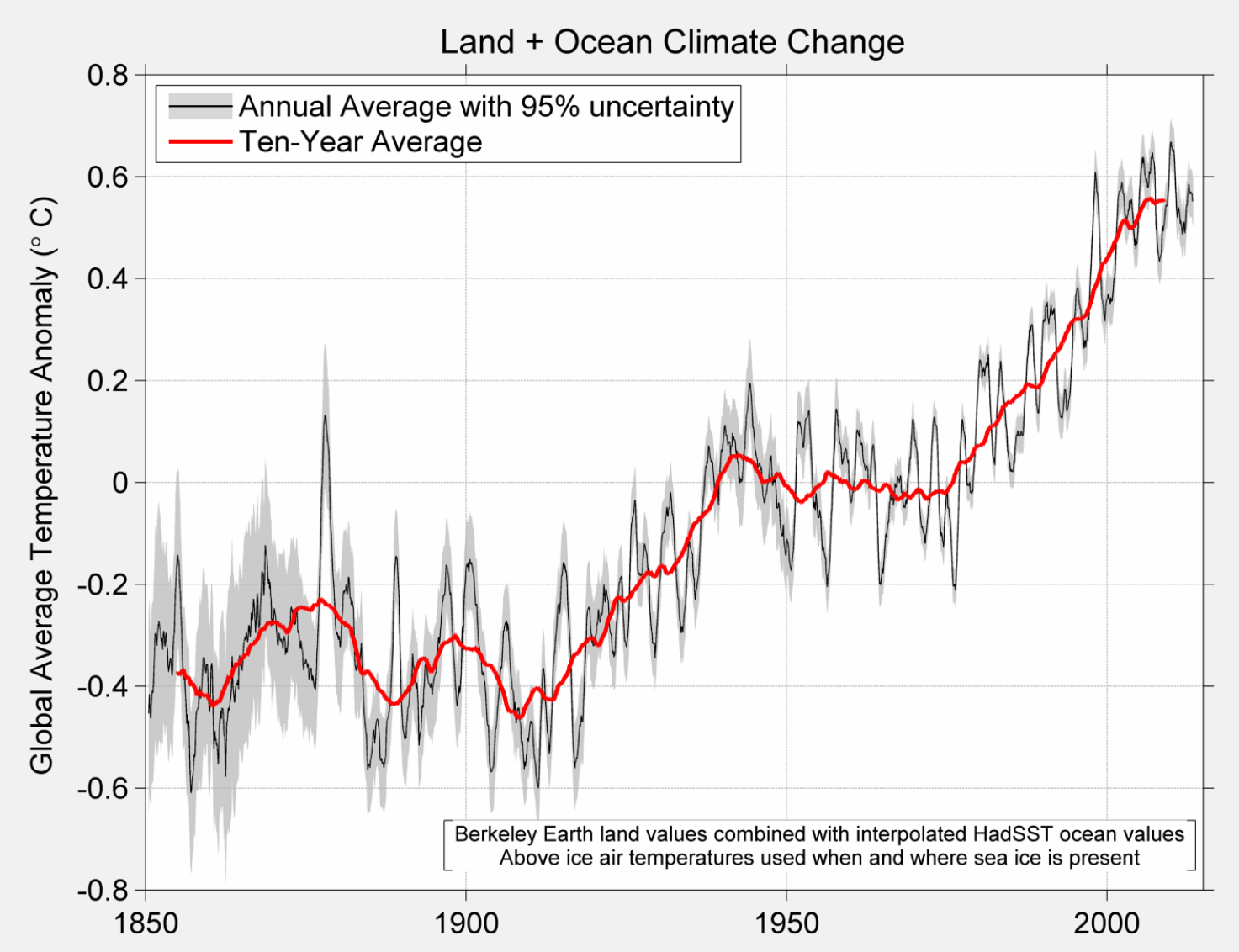
Outline

- What is the warming debate in climate science?
- What are changepoints?
- The story of a statistical solution
- Some frustrations

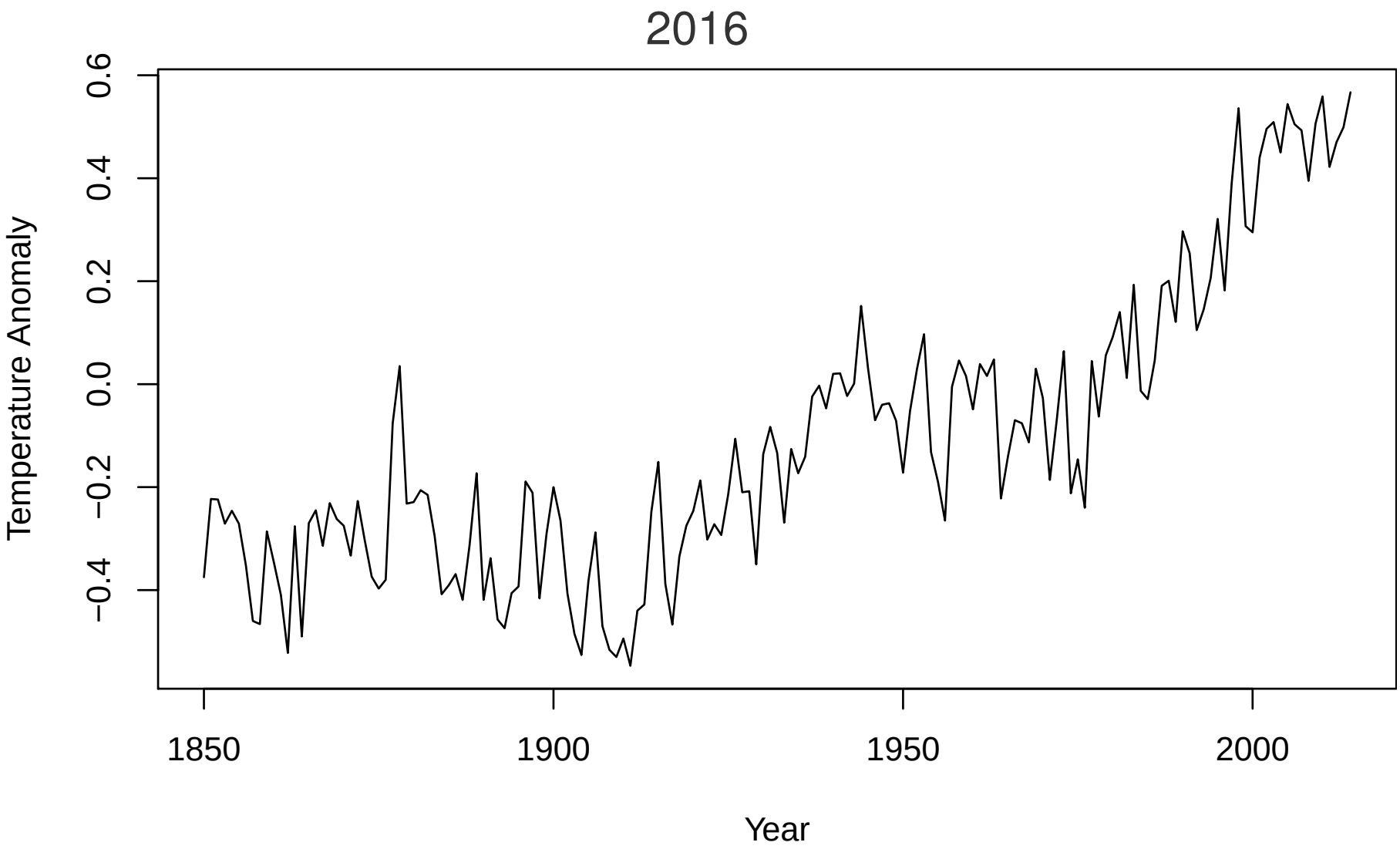
Where my interest began



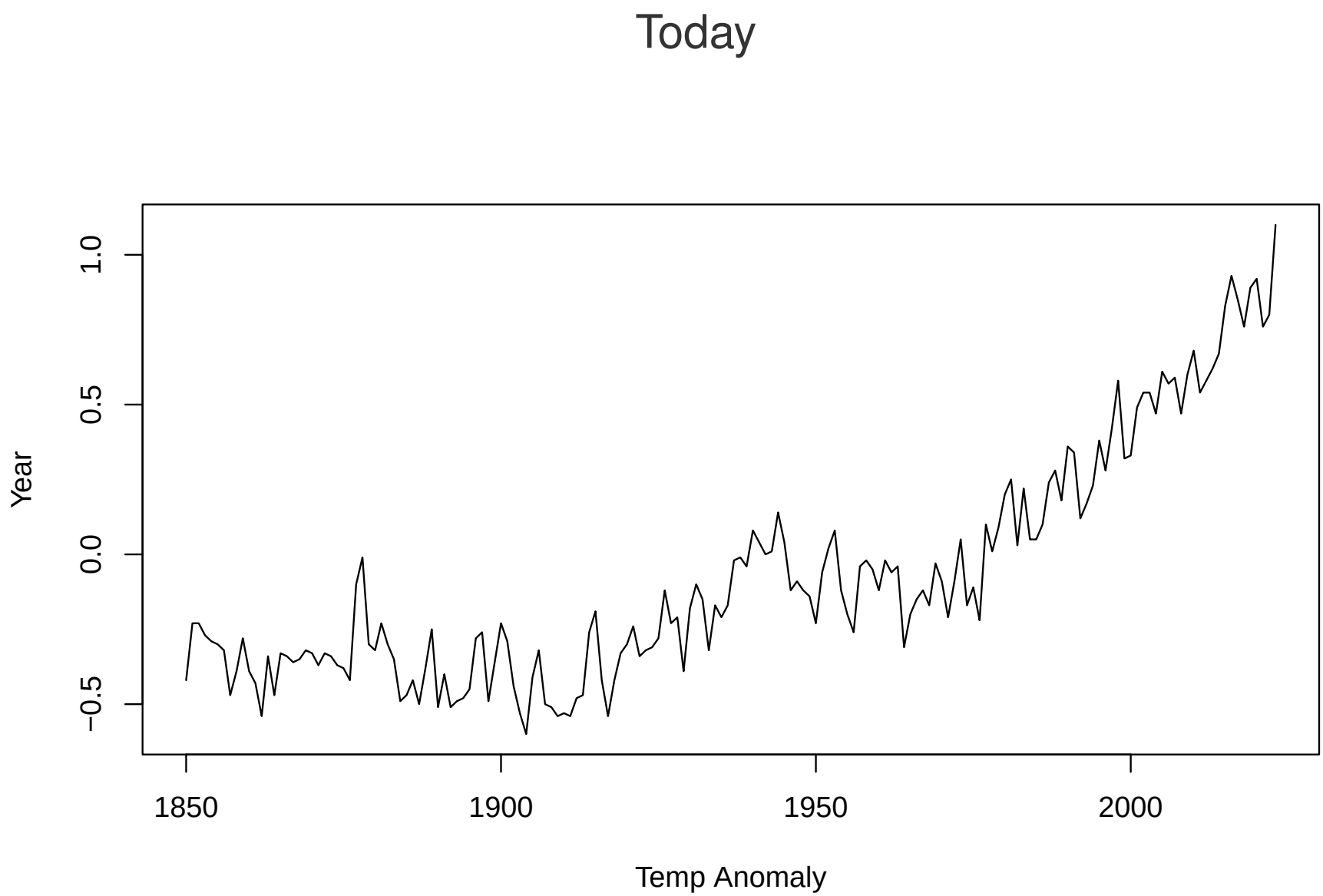
Global Mean Temp 2016



Global Mean Temp



Global Mean Temp



Where is Statistics?

CLEMSON UNIVERSITY

Is climate change speeding up? Here's what the science says.

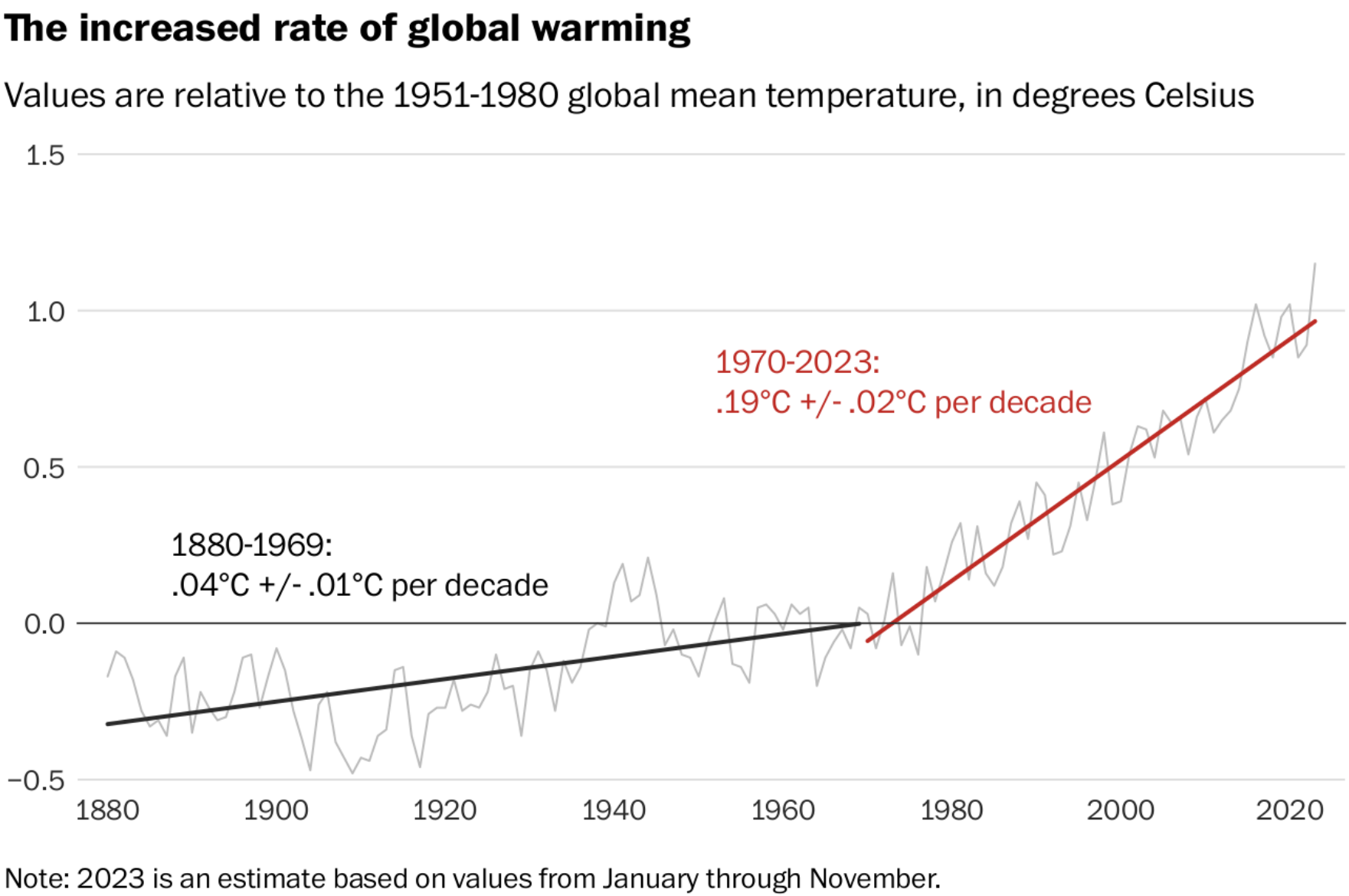
This year's record temperatures have some scientists concerned that the pace of warming may be accelerating. But not everyone agrees.

By Chris Mooney and Shannon Osaka

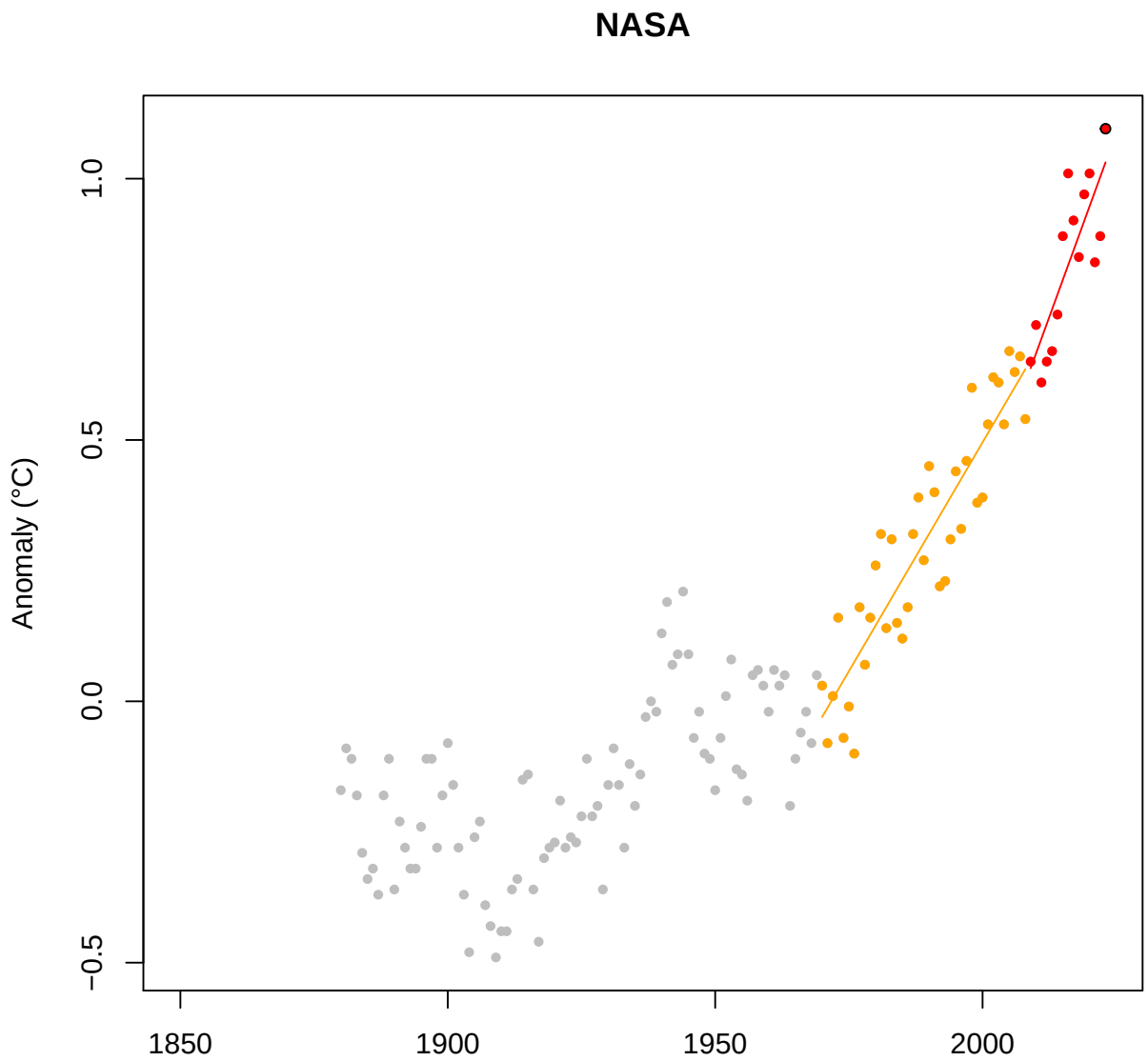
December 26, 2023 at 6:30 a.m. EST

For the past several years, a small group of scientists has warned that sometime early this century, the rate of global warming — which has remained largely steady for decades — might accelerate. Temperatures could rise higher, faster. The drumbeat of weather disasters may become more insistent.

Where is Statistics?



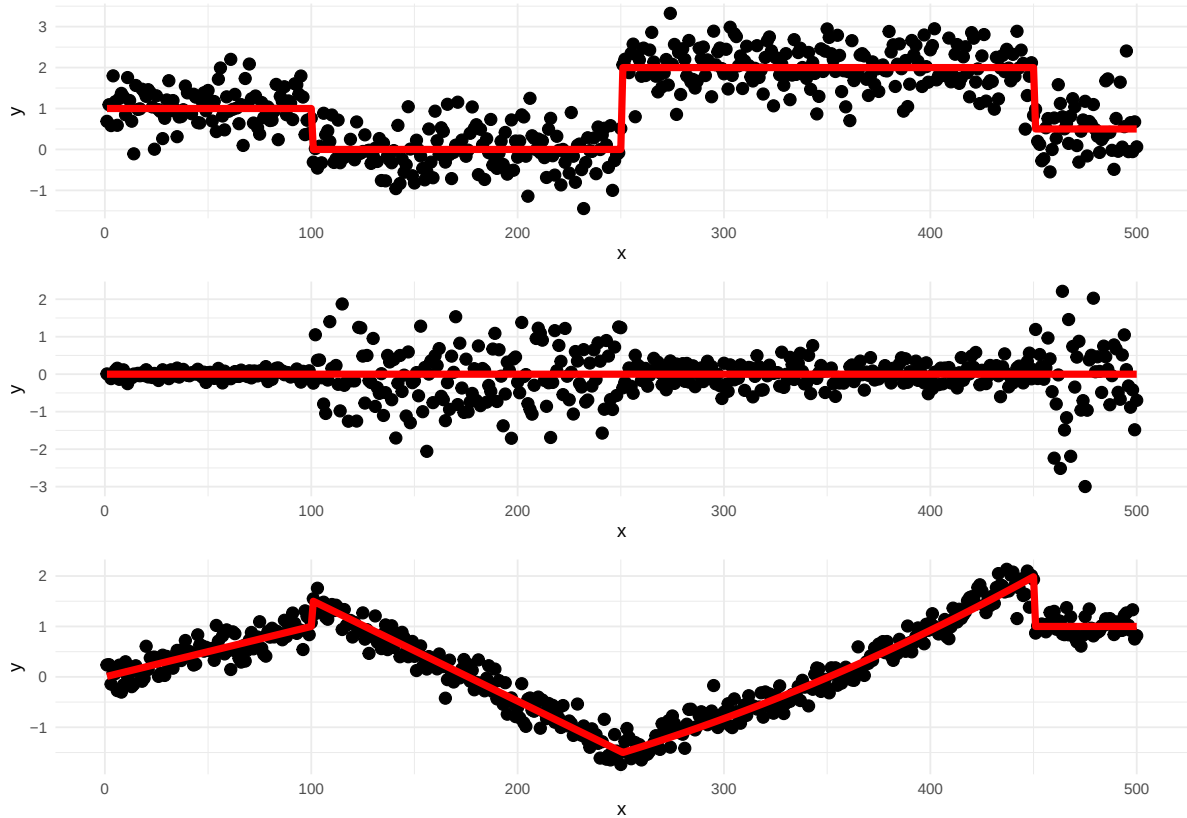
Where is Statistics? NYT



What are changepoints?

For data $y_1, \ldots, y_n$, if a changepoint exists at τ, then $y_1, \ldots, y_\tau$ differ from $y_{\tau+1}, \ldots, y_n$ in some way.

There are many different types of change.

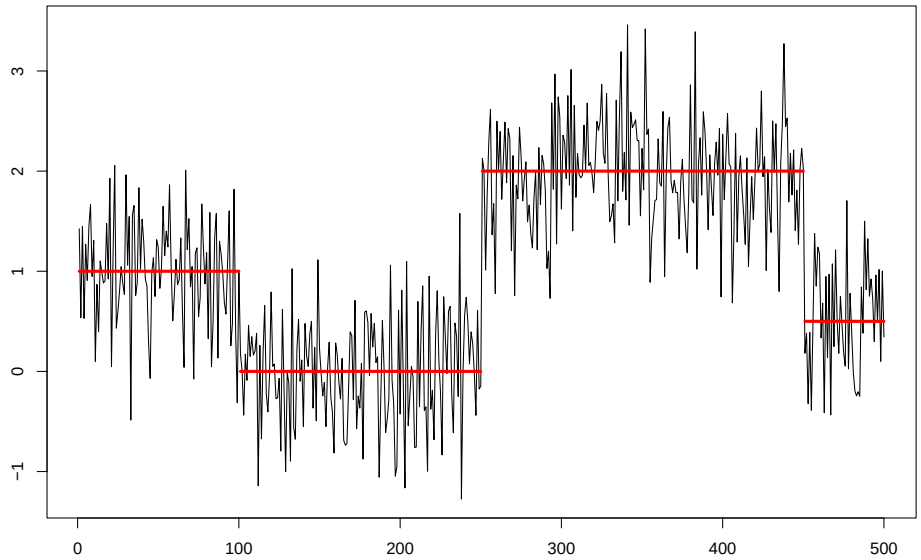


Change in mean

Assume we have time-series data where

$$Y_t|\theta_t \sim \mathrm{N}(\theta_t, 1),$$

but where the means, θ_t, are piecewise constant through time.



Inferring Changepoints

We want to infer the number and position of the points at which the mean changes. One approach:

Likelihood Ratio Test

To detect a single changepoint we can use the likelihood ratio test statistic:

$$LR = \max_{\tau}\{\ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n})\}.$$

We infer a changepoint if $LR > \beta$ for some (suitably chosen) β. If we infer a changepoint its position is estimated as

$$\tau = \arg\max\{\ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n})\}.$$

This can test can be repeatedly applied to new segments to find multiple changepoints.

Likelihood Ratio Tests

Define m to be the number of changepoints, with positions $\boldsymbol{\tau} = (\tau_0, \tau_1, \ldots, \tau_{m+1})$ where $\tau_0 = 0$ and $\tau_{m+1} = n$.

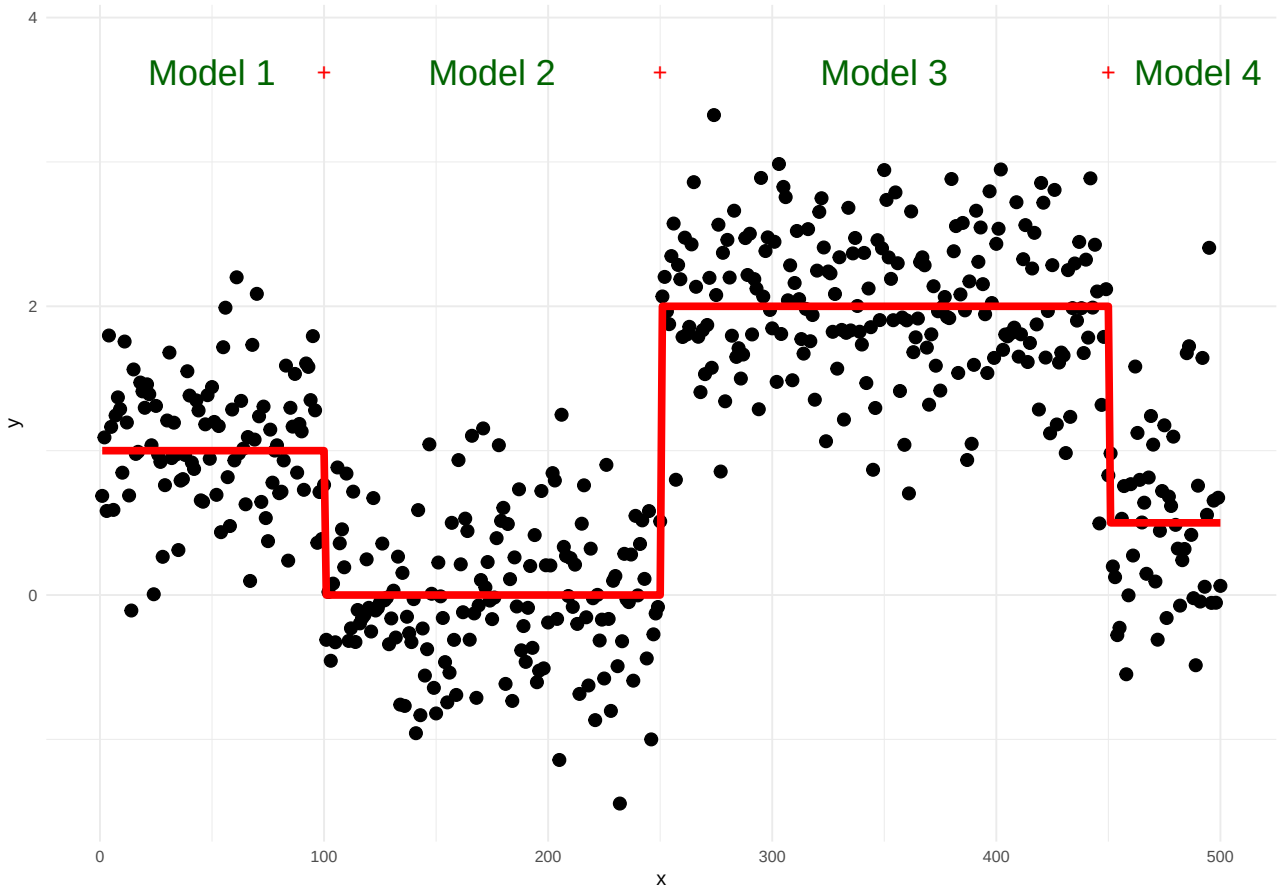
Then one application of the Likelihood ratio test can be viewed as

$$\min_{m \in \{0,1\}, \boldsymbol{\tau}} \left\{ \sum_{i=1}^{m+1} \left[-\ell(y_{\tau_{i-1}:\tau_i}) \right] + \beta m \right\}$$

Repeated application is thus aiming to minimise

$$\min_{m, \boldsymbol{\tau}} \left\{ \sum_{i=1}^{m+1} \left[-\ell(y_{\tau_{i-1}:\tau_i}) \right] + \beta m \right\}$$

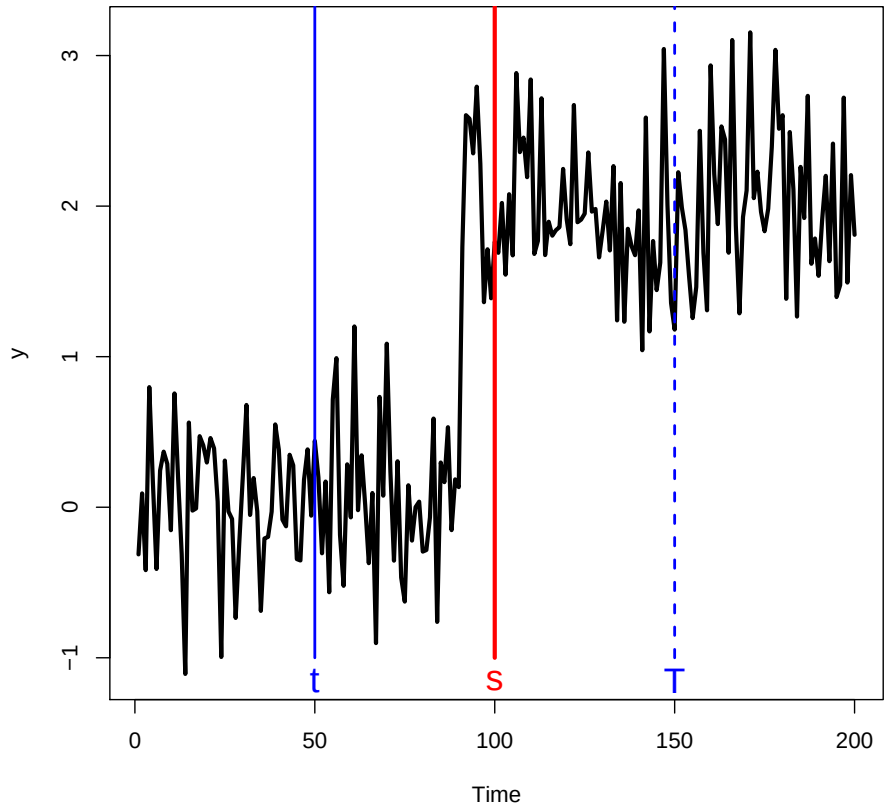
Problem



- How many changes?
- Where are the changes? 2^{n-1} possible solutions!

PELT in a nutshell

- Dynamic programming allows us to only worry about the location of the *last* change.
- Pruning means that as we go through the data we are smart about which locations are potential last change locations.

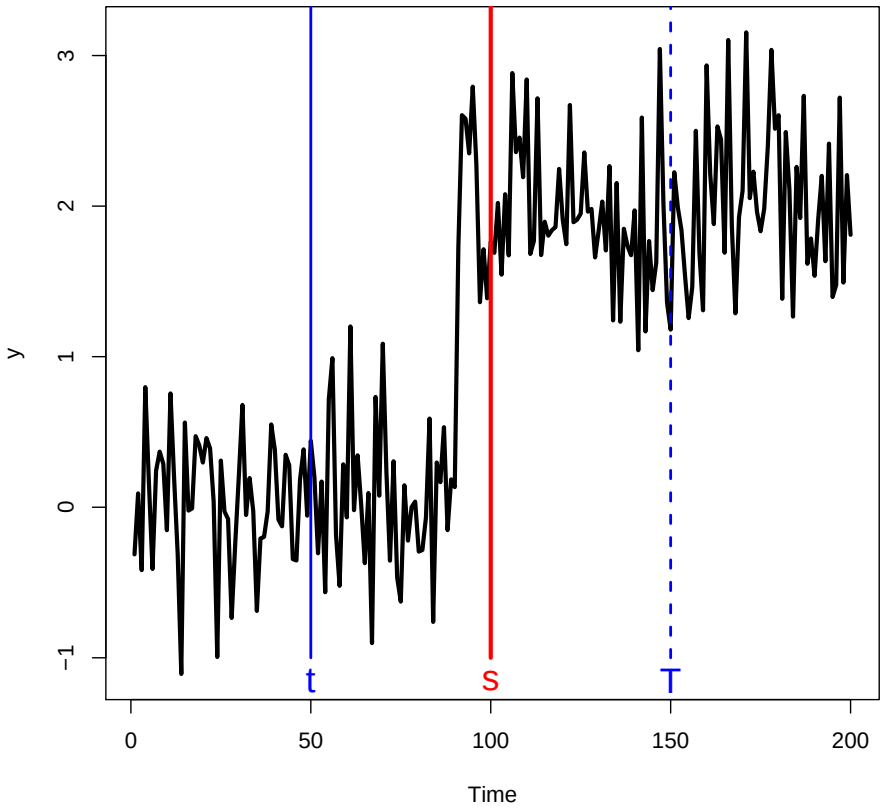


PELT: Pruning

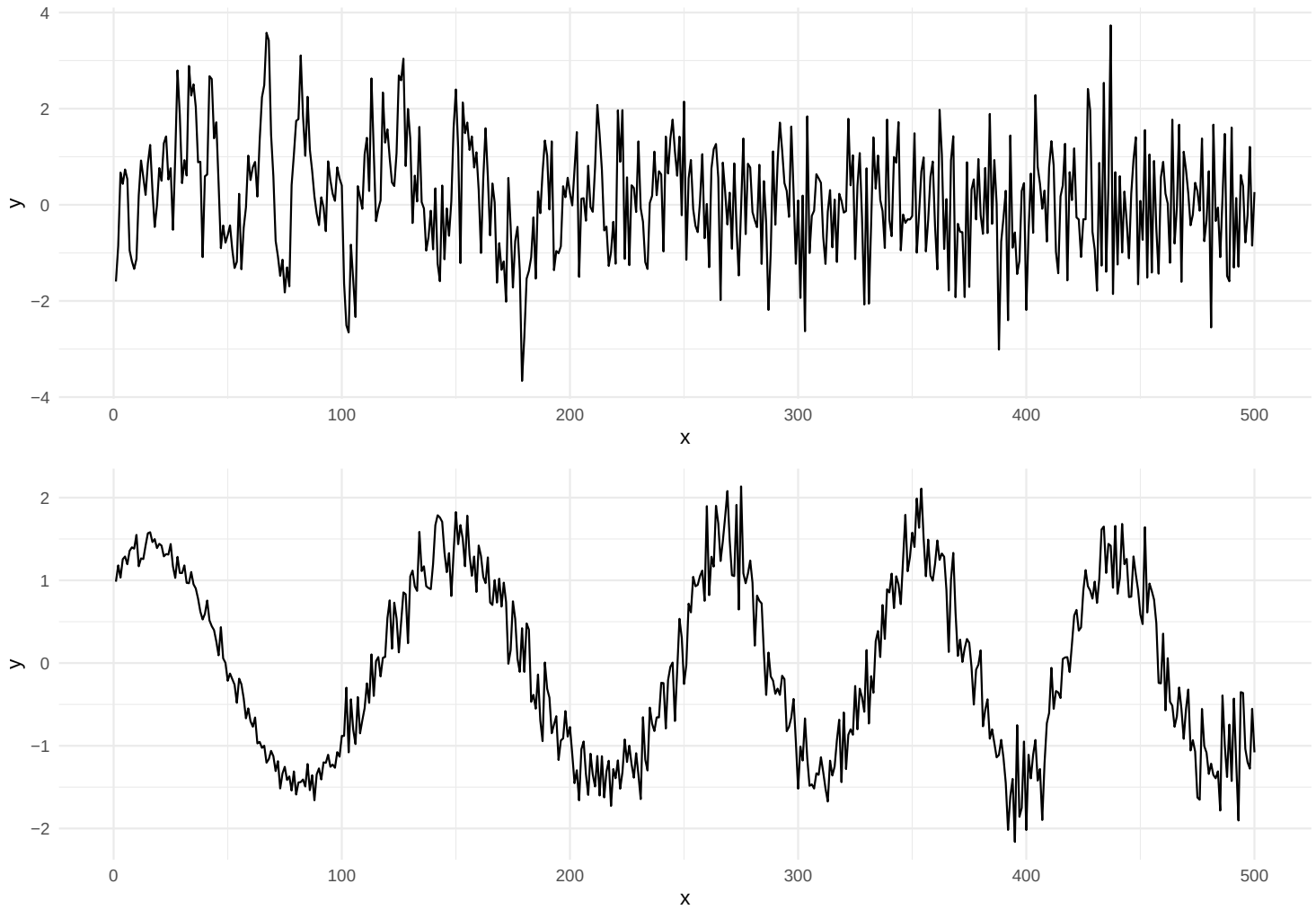
Let $0 < t < s < T$, if

$$F(t) + \mathcal{C}(y_{(t+1):s}) < F(s)$$

then at any future time $T > s$, t can never be the optimal last changepoint prior to T.
We can prove that, under certain regularity conditions, the expected computational complexity will be $\mathcal{O}(n)$.



More complicated changes



Interesting Math Challenges

- Optimization
- Model development
- Theoretical properties
- Limits of detection
- Small sample and large sample in the same data

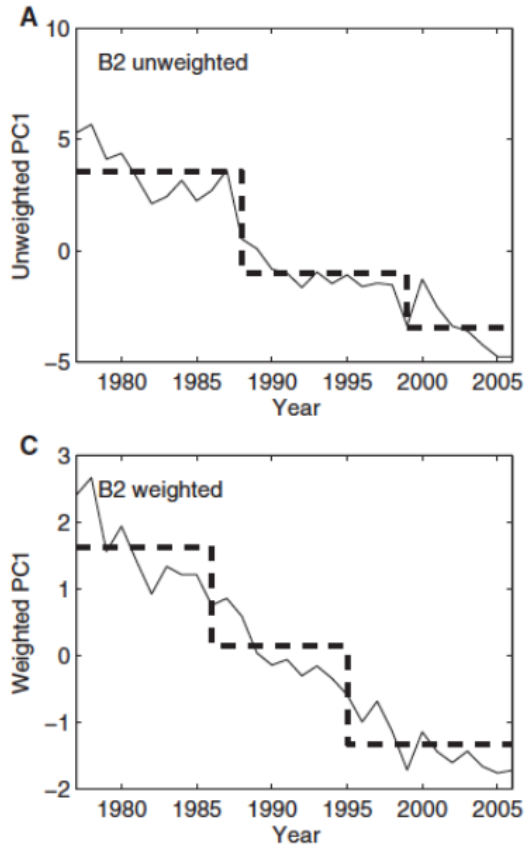


Key practical challenges

- Model building (assumptions)
- What can change?
- What fitting algorithm to use?
- Penalty selection
- Dealing with outliers

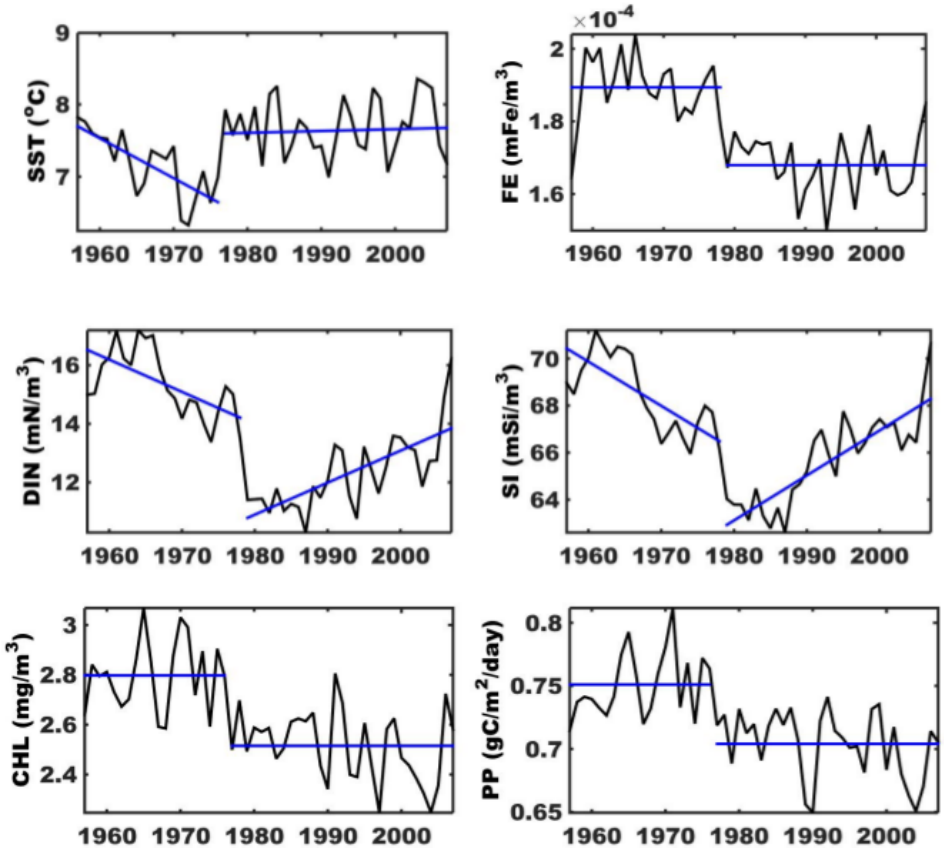
Tooling not enough

- From a publication in Marine Ecology (not the only one)
- Used the Rodionov (2004) method.
- Very popular but cannot deal with trend or autocorrelation.



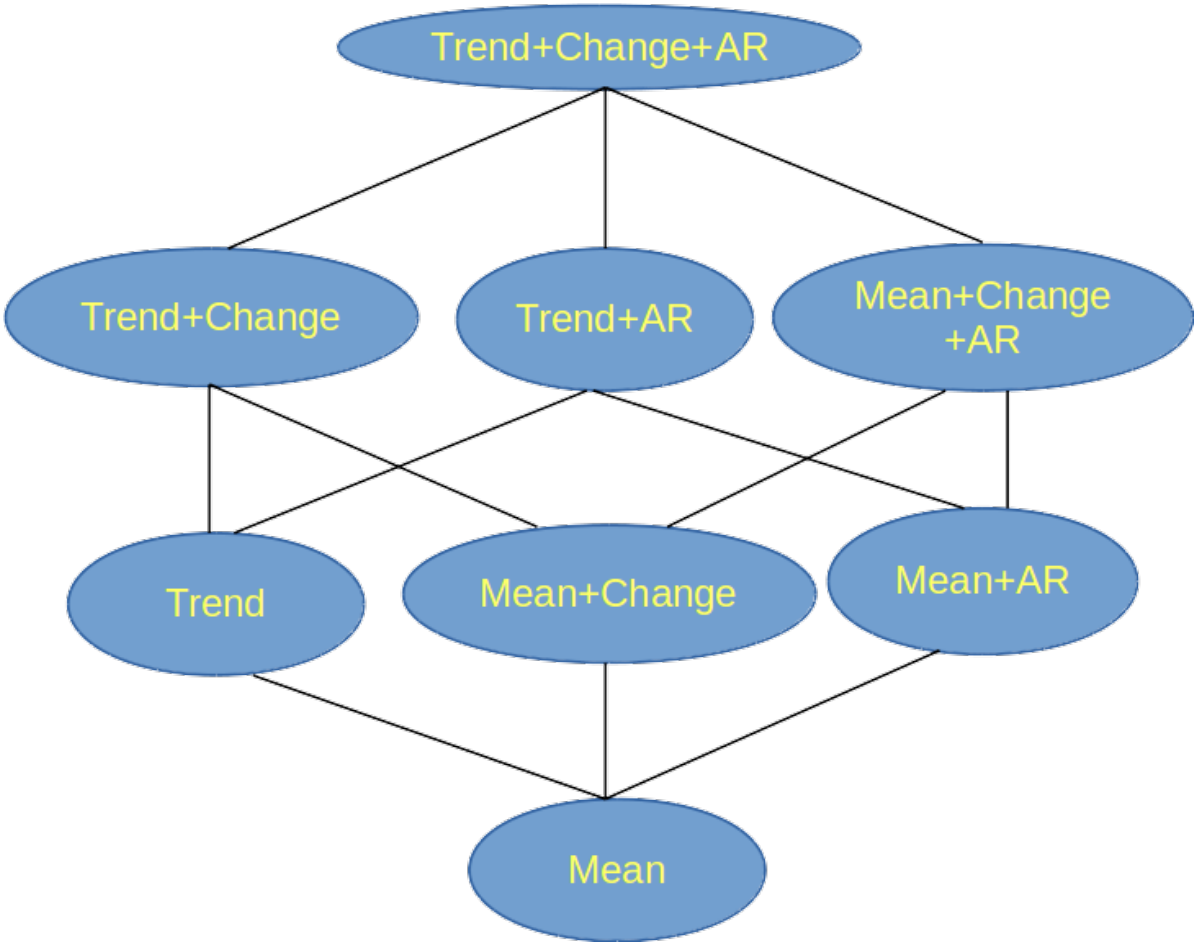
Beaulieu et al. 2015

- potentially hundreds or thousands of series
- no time to consider the format of change for each
- need to include both the potential for trends and also red noise (autocorrelation).

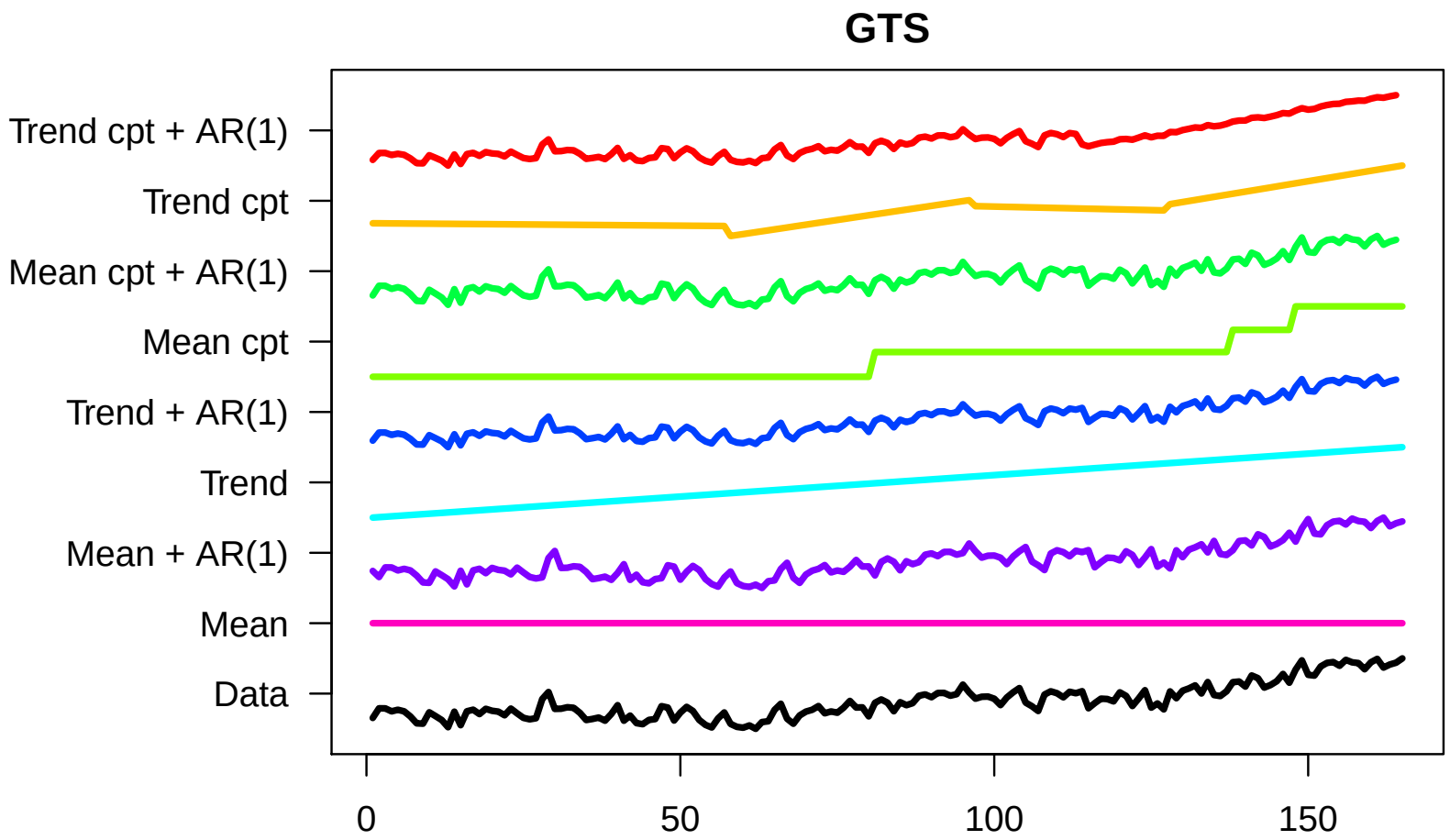


Model Selection

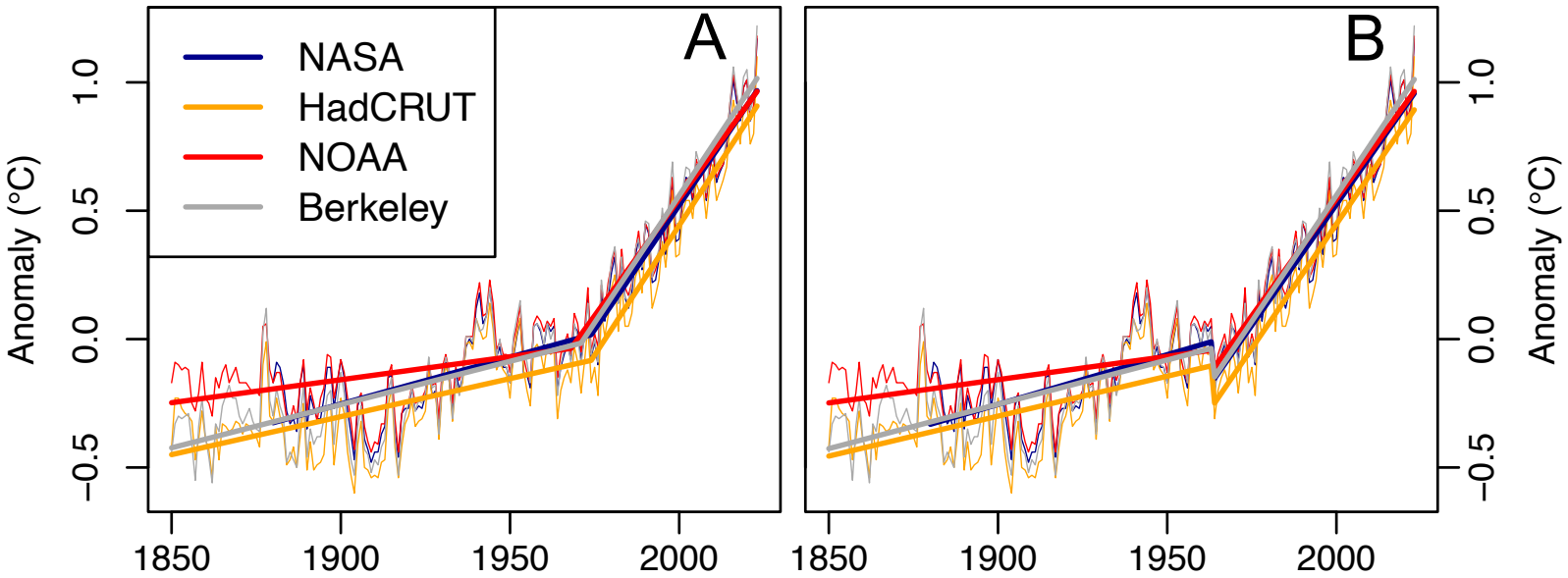
`EnvCpt`: select the most parsimonious but accurate model for the data. Simple to extend with other types of models.



GMST All Fits



Join-pin Models



Join-Pin IID

Fearnhead, Maidstone, Letchford, JCGS (2019)

$$Y_t = \theta_i + \frac{\theta_{i+1} - \theta_i}{\tau_{i+1} - \tau_i}(t - \tau_i) + Z_t$$

where $Z_t \sim \mathrm{N}(0, \sigma^2)$.

Implemented in the `cpop` R package.

Join-Pin AR(p)

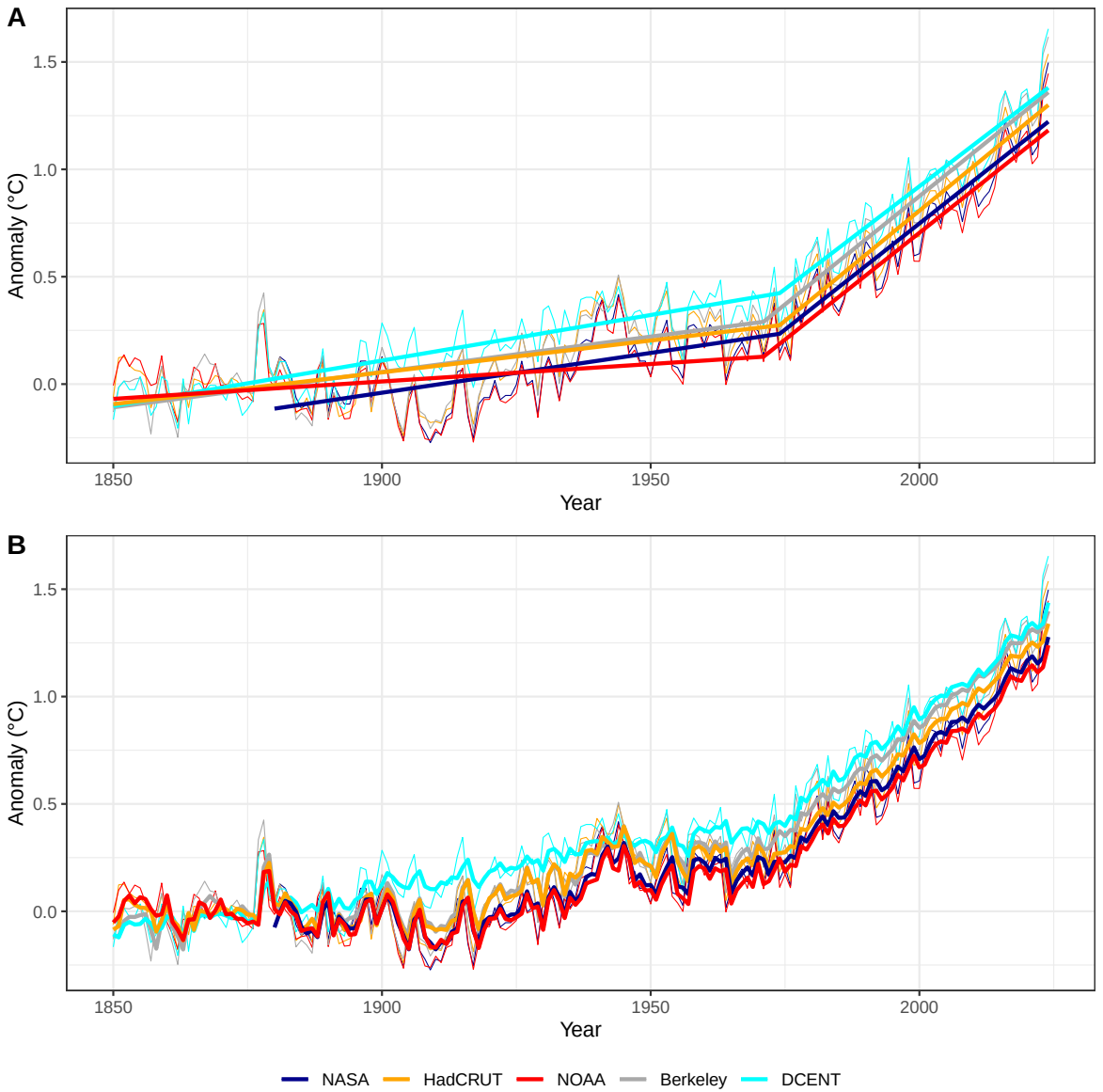
$$Y_t = \theta_i + \frac{\theta_{i+1} - \theta_i}{\tau_{i+1} - \tau_i}(t - \tau_i) + Z_t$$

but now Z_t is AR(p), $Z_t = \phi_1 Z_{t-1} + \phi_2 Z_{t-2} + \ldots + \phi_p Z_{t-p} + \epsilon_t$.

Challenge:

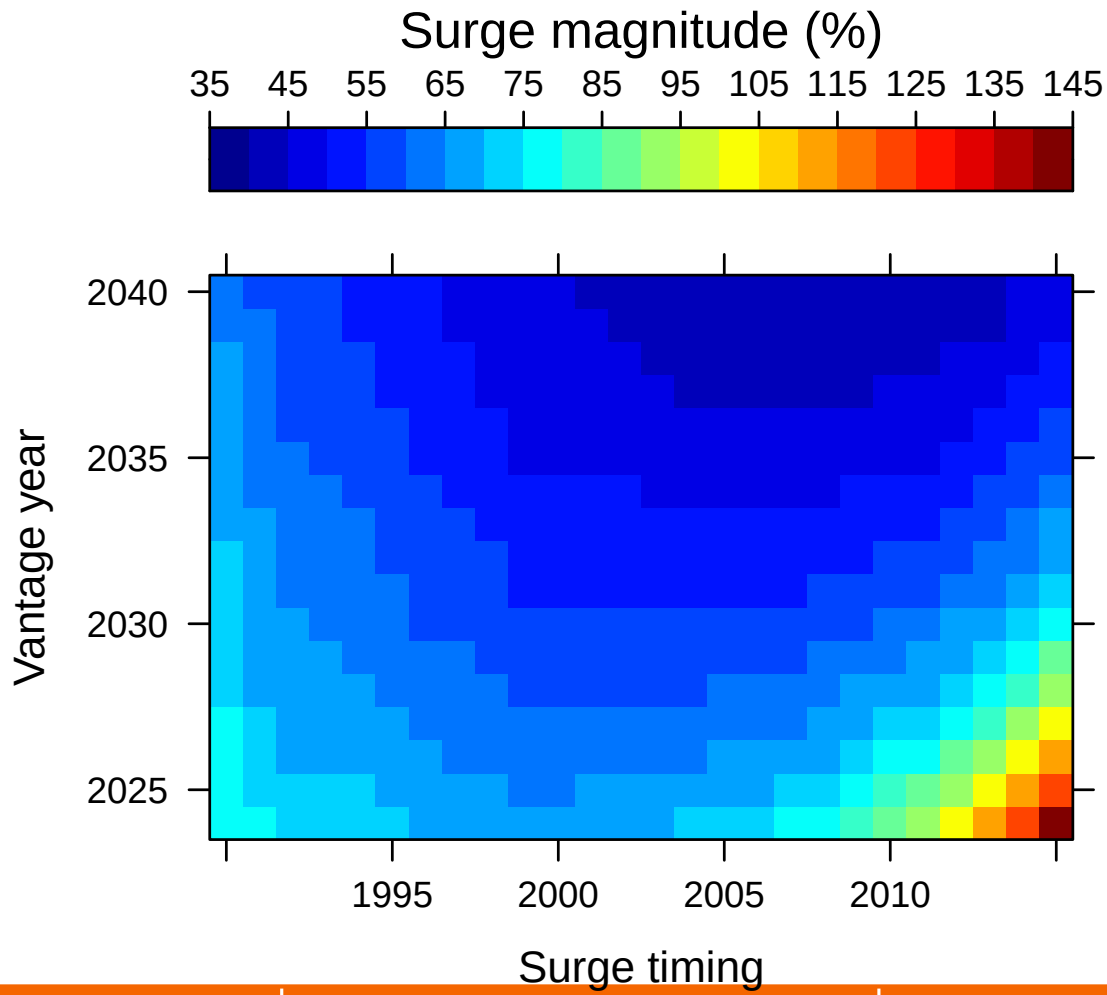
- Definition of join-pin
- Fitting AR parameter across segments:
  - Fixed we use EM algorithm
  - Varying we can embed in PELT

Join-pin Models



So what?

What would we need to see (in the statistically preferred model).



The fallout

Global warming is NOT surging, scientists say - despite record-breaking temperatures

- **Scientists find no evidence for a 'surge' in global temperatures**
- **READ MORE: 2023 was officially the hottest year on RECORD**

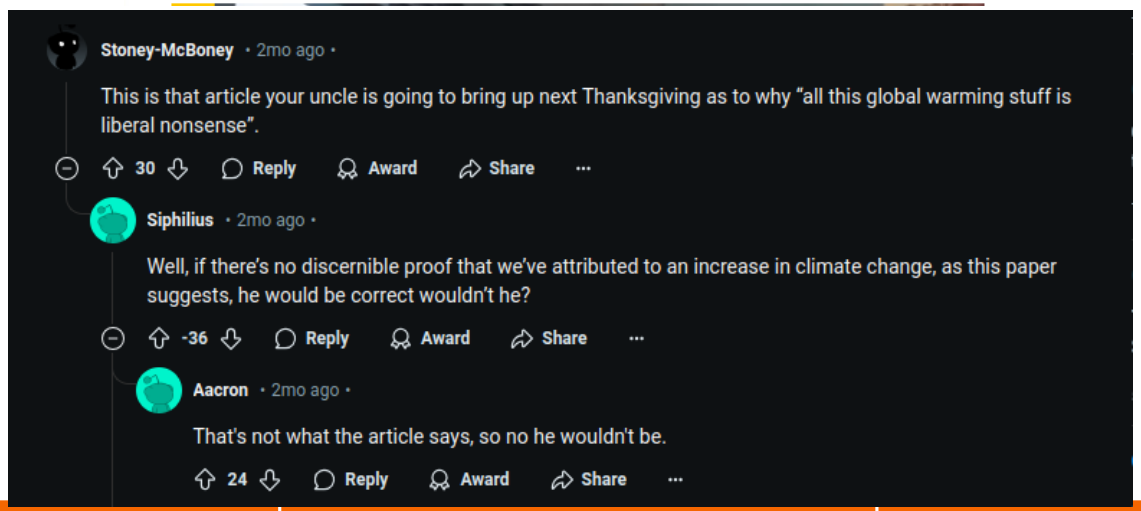
By JONATHAN CHADWICK FOR MAILONLINE
PUBLISHED: 14:01, 15 October 2024 | **UPDATED:** 07:52, 16 October 2024

Share 138 shares 407 View comments

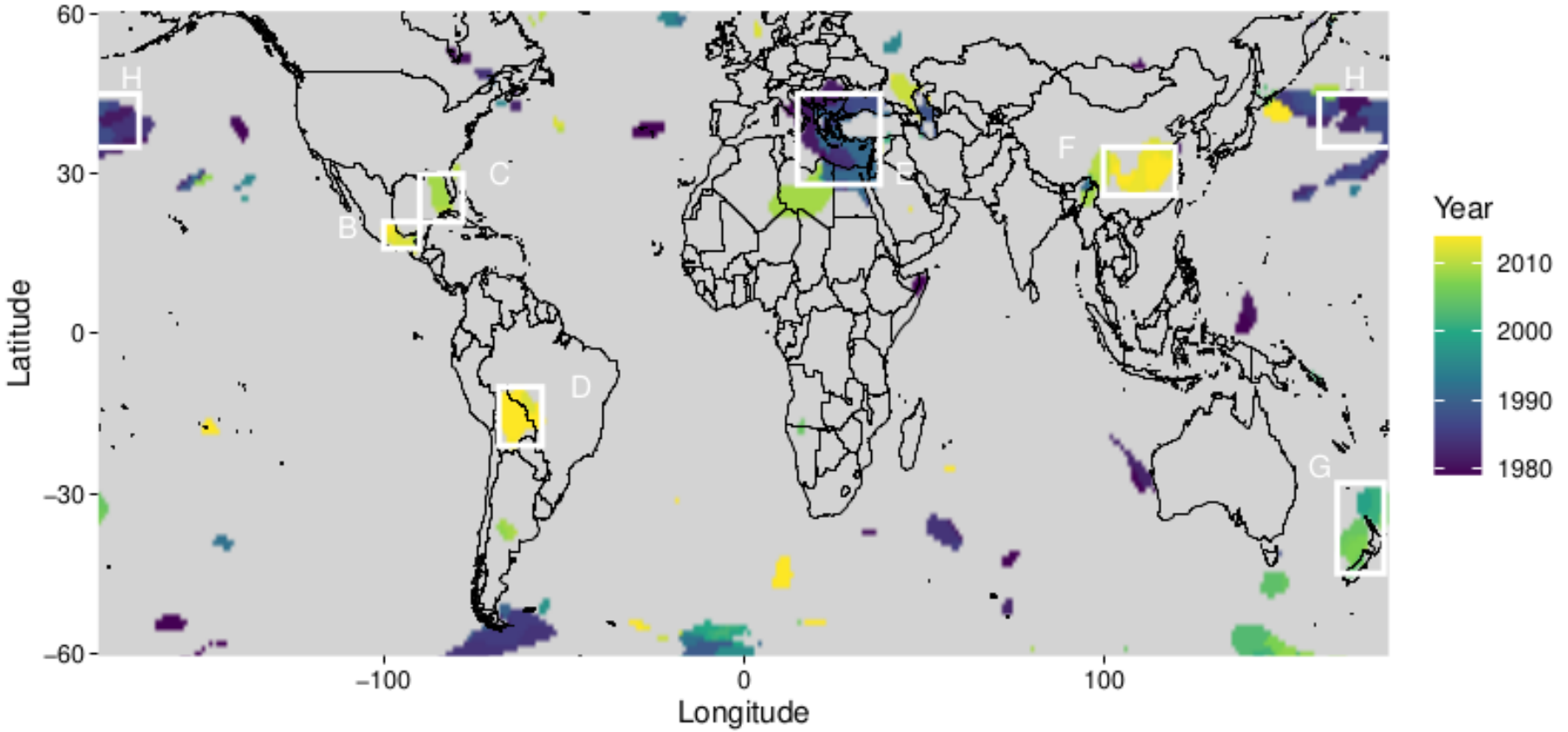
From the **UK's hottest day** to the **hottest year on record globally**, there's no doubt some worrying temperature records have been broken in recent years.

Many people think the rate of **global warming** has dramatically accelerated or 'surged' over the past 15 years – and is a cause of more extreme weather.

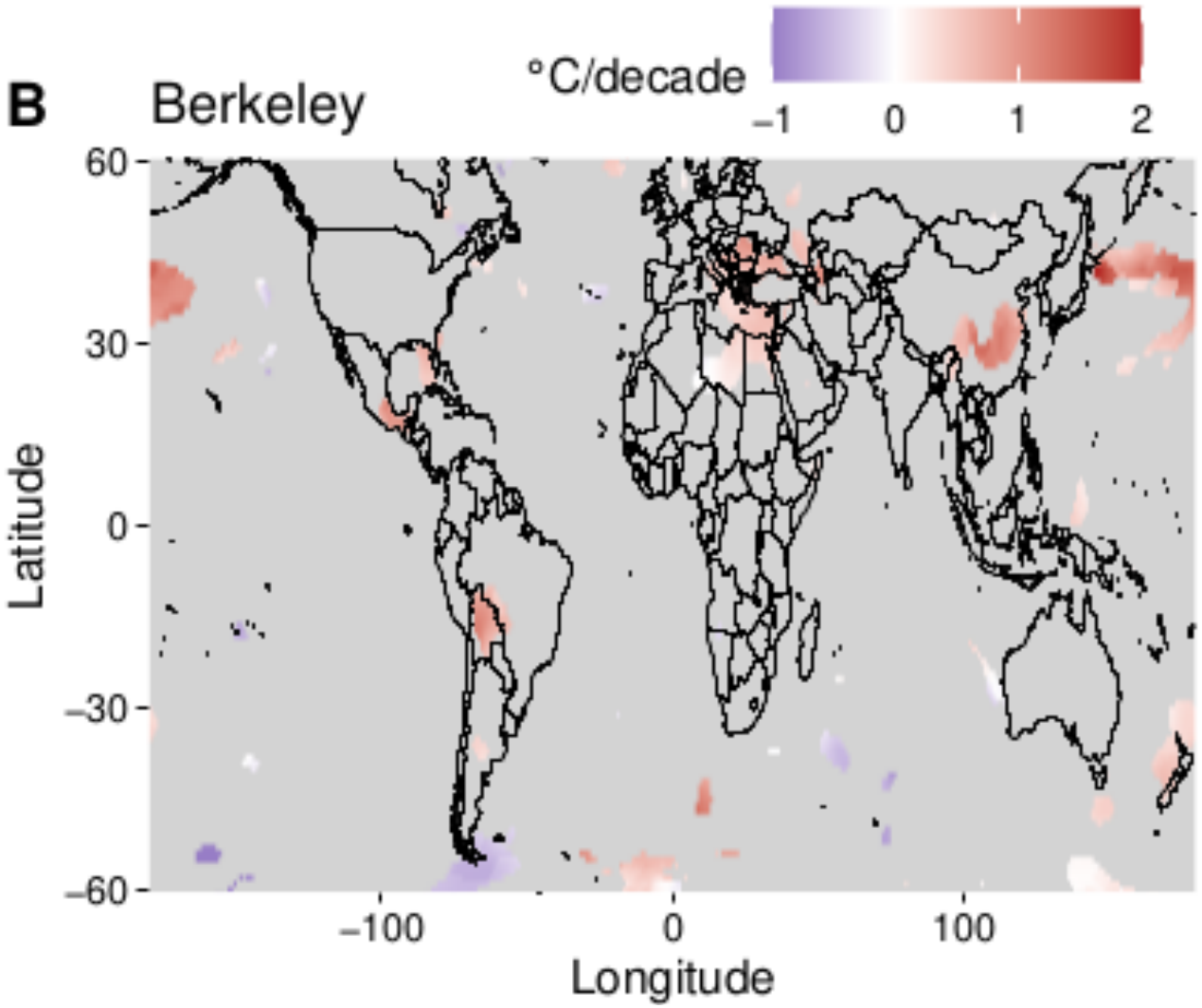
But a new study says there is not any statistical evidence for this so-called 'surge' or 'leap'.



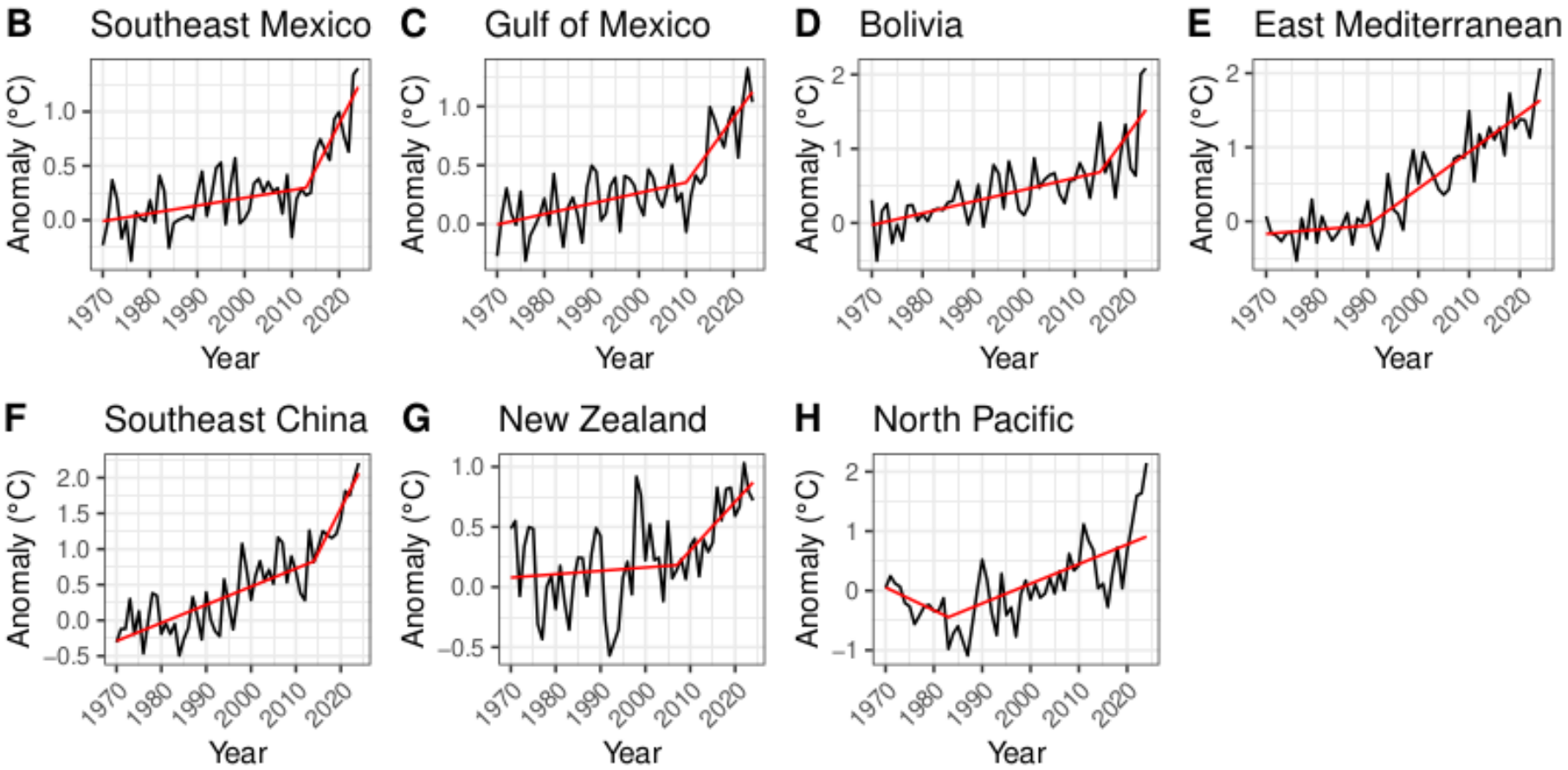
Extension to gridded



Extension to gridded

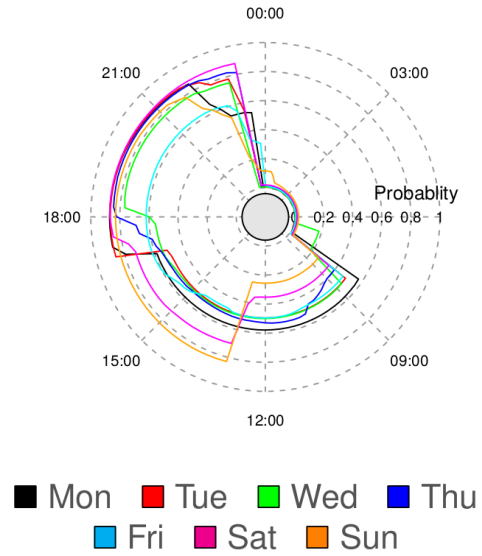


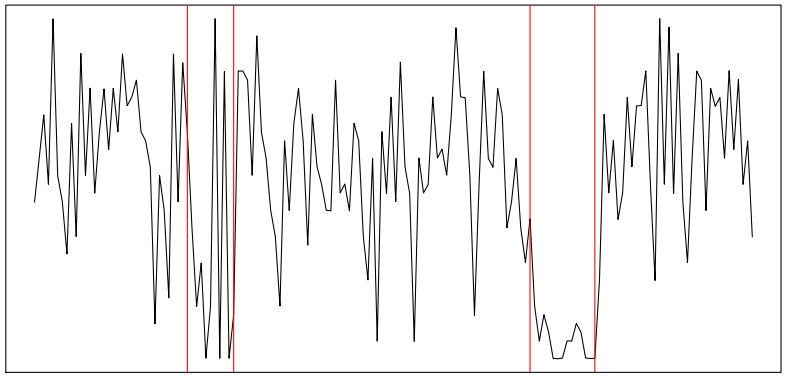
Extension to gridded



Other types of data

- Passively monitor patients at home
- Desire to have early intervention before crisis point is reached

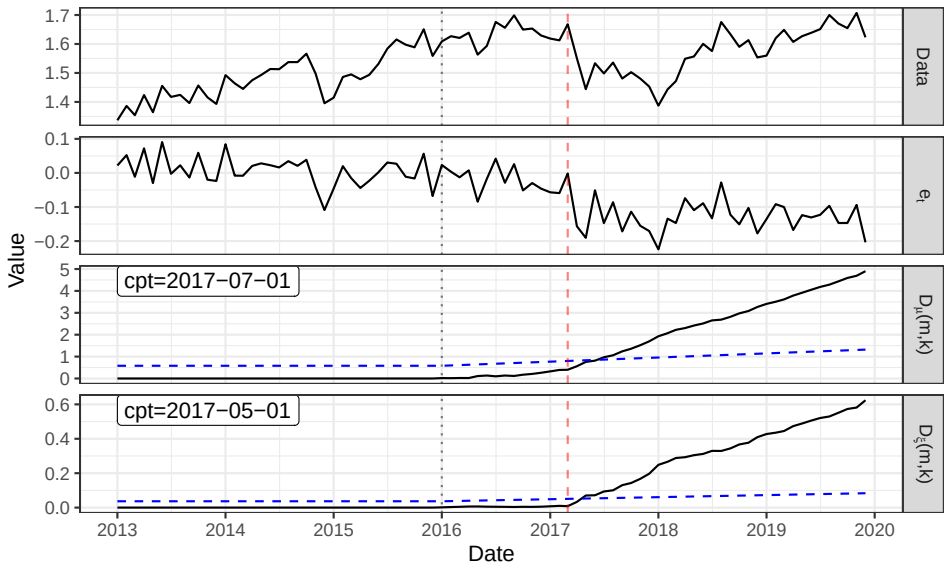


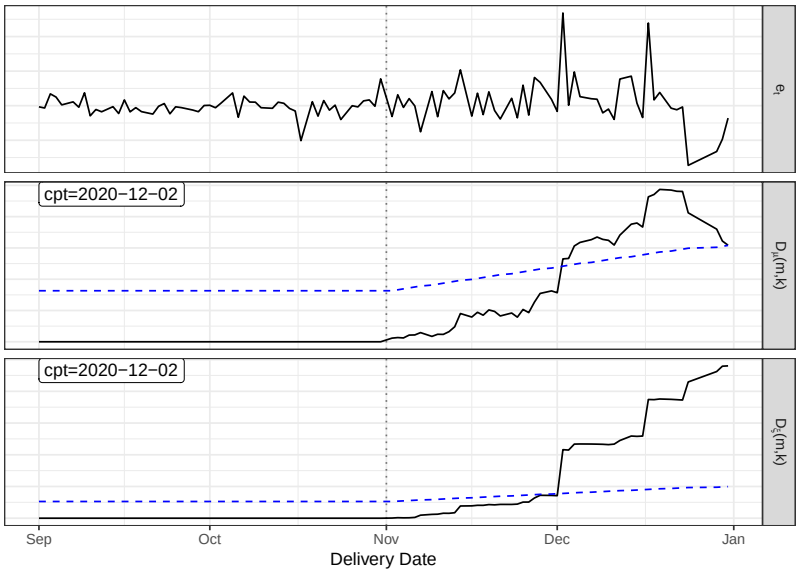


- View data as circular, e.g., each day.
- Developed circular changepoint detection to detect changes in activity levels within a day
- Changes in the detected routine signal early intervention

Concept drift

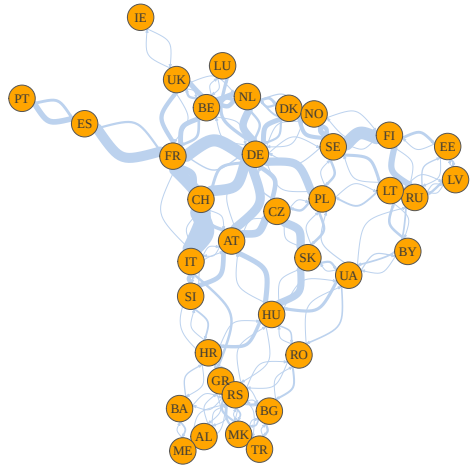
Monitor models for concept drift

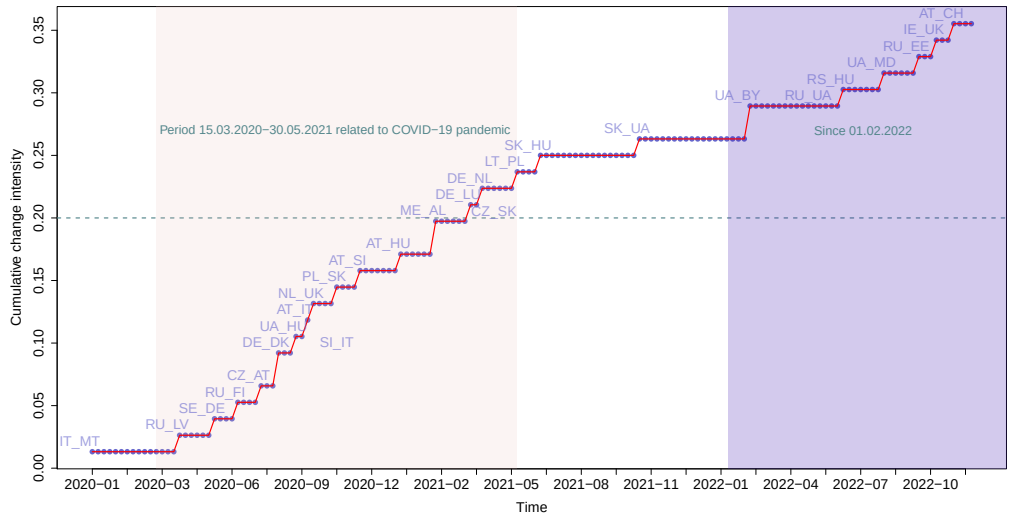




Networks

Concept drift in time series on networks





Summary

- Detecting and documenting changepoints improves analyses . . .
- . . . and can help answer pertinent questions in different domains
- Extending to more complex model structures is interesting statistically
- Extending to monthly/grid point is interesting climatically

- I enjoy working with different disciplines . . .
- . . . as often it sparks my next research challenge.

Papers discussed

Preprints of all available at: www.lancs.ac.uk/~killick/pub.html

PELT: https://doi.org/10.1080/01621459.2012.737745

Model Choice: https://doi.org/10.1175/JCLI-D-17-0863.1 & https://doi.org/10.1002/qre.2712

LMvsCpts: https://doi.org/10.1007/s11222-017-9731-0 & https://doi.org/10.1002/env.2568

Warming Surge: https://doi.org/10.1038/s43247-024-01711-1

Regional Surge: https://doi.org/10.1038/s41467-026-77514-z

Climate Dynamics

